Multi-Horizon Joint-Embedding Predictive Architectures and Coordinate Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Reconstructive time-series anomaly detectors score raw observations, making them sensitive to benign sensor jitter and drift. Joint-Embedding Predictive Architectures (JEPAs) forecast normalized latent window summaries, yet collapsing coordinates into one scalar residual allows drifting coordinates to inflate thresholds. We introduce two multi-horizon circuit-breaker models: Laminar-JEPA, a four-horizon, 224-dial panel combining sub-horizon magnitudes, drift, fluctuation, endpoint differences, and 32 onset relays with optional nominal weight-space quenching; and Laminar-JEPA-Lite, a distilled three-horizon, 128-dial position-plus-drift detector using a parallel single-normalization predictor and channel-gated spectral splitting with zero weight edits. Laminar-JEPA-Lite cuts parameters by 35 to 52 percent and accelerates inference by 2.43 times. On the 33-stream development benchmark, Laminar-JEPA and Laminar-JEPA-Lite hold autonomous causal false-positive rate (FPR) at 1.3 and 1.0 percent versus 22.2 to 24.3 percent for reconstructors, reaching buffered point-wise F1 of 0.244 and 0.242 and seven-dataset macro F1 of 0.133 and 0.171. On the 19-stream residual cache, the 224-dial panel lifts causal F1 from 0.013 to 0.071 over scalar collapse, and a 160-dial subset reaches 0.098. Across 118 untouched held-out streams under disjoint checkpoint selection, Laminar-JEPA-Lite lifts causal F1 from 0.074 to 0.085 and event recall from 49.5 to 67.0 percent over scalar multi-horizon scoring while remaining positive across all 12 leave-one-dataset-out folds on 141 streams, whereas quenching lowers four-horizon causal FPR across seven of nine datasets. Calibration precision depends on autocorrelation-adjusted sample size; causal ramp detection vanishes at four context windows, and single-sensor sensitivity raises false alarms.**

***Index Terms*—Anomaly detection, Multivariate time series, Representation learning, Self-supervised learning, Statistical process control.**

I. Introduction

Telemetry monitors track dozens of channels simultaneously [1]–[4]. Because labeled faults are scarce, a streaming detector must learn nominal behavior from healthy history alone, flag anomalies promptly, and keep false alarms low during normal operation.

Why Observation-Space Reconstructors Suffer High False-Alarm Rates: Reconstructors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] score raw sensor error. Because raw telemetry mixes jitter, thermal drift, and disparate channel scales, summing errors across 25–55 channels produces heavy-tailed spikes. Calibrated on healthy windows via Streaming Peaks-Over-Threshold (SPOT) [8], reconstructors fire during 22%–31% of later healthy data; a rate-matched coin flip catches 60.0% of anomaly events, exceeding TimesNet's 58.8%.

The Bottleneck of Scalar Score Aggregation: Joint-Embedding Predictive Architectures (JEPAs) [9], [10] compress each window into a 32-dimensional normalized summary and forecast future summaries in latent space, cutting nominal false alarms to 0.6%–2.4%. Yet scalar latent detectors still sum all coordinate errors against one threshold: if one or two coordinates shift slightly between training and calibration (*hijacker dials*), they inflate the global threshold and mask faults on the other 30 coordinates (causal Point-F1 0.050–0.055, delay 138–154 steps).

*Multi-Horizon Coordinate Circuit Breakers (*Laminar-JEPA *and* Laminar-JEPA-Lite*):* We pair multi-horizon JEPA forecasting with calibrated Boolean circuit-breaker panels in two tiers (Fig. 1): (1) *Laminar-JEPA* maps context and target windows into 32D layer-normalization (LayerNorm) [11] summaries with resolvent residual whitening, expands each coordinate's four-horizon error trajectory into $7 \times 32 = 224$ dials plus 32 right-edge onset relays, and applies optional nominal split-gated weight-space quenching; and (2) *Laminar-JEPA-Lite*, distilled from a layer-by-layer activation and component autopsy, replaces sequential four-horizon rollout and post-hoc weight editing with a parallel three-horizon single-LayerNorm predictor, channel-count-gated spectral splitting ($K \geq 25$), and $4 \times 32 = 128$ position-plus-drift breakers under a single nominal quorum, cutting parameters by 35%–52% and accelerating windowed inference by $2.43\times$.

A. Contributions

We separate evaluation into a 33-stream development cohort (plus a 19-stream residual cache) and a disjoint 118-stream untouched held-out cohort within a 141-stream archive across 12 datasets under disjoint checkpoint selection (Section V-B):

1) *Flagship and distilled panels:* On the 33-stream benchmark, *Laminar-JEPA* and *Laminar-JEPA-Lite* reach buffered Point-F1 of 0.244 and 0.242 and causal event recall (ER) of 67.3% and 65.0% at 1.3% and 1.0% held-out false-positive rate (FPR), with *Laminar-JEPA-Lite* leading 7-dataset macro Point-F1 (0.171 vs. 0.133). On the 19-stream

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Andre Williams, Imadeldin Mahgoub, and Hamid Akbarian are with the Department of Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431 USA (e-mail: andrewilliam2018@fau.edu; mahgoubi@fau.edu; hakbaria@fau.edu).

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cache, the 224-dial panel lifts causal Point-F1 from 0.013 to 0.071 over 1D collapse ($p = 0.0023$), and a 160-dial subset reaches 0.098. Across 118 untouched streams ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$), *Laminar-JEPA-Lite* raises causal Point-F1 from 0.074 to 0.085 ($p = 0.0034$; $+0.025$ over TS-JEPA, $p = 1.6 \times 10^{-4}$), lifts causal ER from 49.5% to 67.0% across $8/9$ datasets (sign-flip $p = 0.012$), and stays positive in all $12/12$ leave-one-dataset-out (LODO) folds on $n = 141$, while nominal quenching lowers 4H causal FPR across $7/9$ datasets ($p = 0.016$).

2) *Autopsy distillation, mechanisms, and latency:* Matched-backbone controls (Table I(c)) isolate LayerNorm tail compression and barycenter drift reduction while showing scalar spectral reweighting is null. Layer activation traces, LayerNorm spillover audits, and a $K \geq 25$ spectral threshold motivate *Laminar-JEPA-Lite*. Across $S \in \{1, 4, 16, 32, 64\}$ on five exploratory streams, both panels reach 2.1–4.3$\times$ the scalar causal Point-F1 at $S = 1, 4$ without establishing cross-domain latency rules.
3) *False-alarm control and contextual faults:* On the 19 non-GHL streams, latent prediction holds autonomous causal FPR at 0.6%–2.4% versus 22.2%–24.3% for reconstructors ($N = 7$ dataset $p = 0.016$). At diagnostic budget $f^* = 1\%$ test FPR, *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% and 56.0% of events versus 21.7% for TimesNet, and reach 0.519–0.532 buffered precision-recall (PR) AUC with 77.5%–87.5% autonomous ER on NASA contextual faults (Tables III–IV, Fig. 4).
4) *Calibration sample size, single-channel branch, and limits:* Proposition 1 bounds horizon-fluctuation concentration $R_{\text{LT}} \in [\frac{1}{M-1}, 1]$, while scalar R_{LT} reweighting is null (ΔF1 $= -0.005$). Sokal autocorrelation (τ_{int} up to 206) and block-bootstrap audits show $N_{\text{eff}} = 300$ does not guarantee 99.9th-percentile precision; interquartile-range (IQR) and Beta-smoothed quantiles remain exploratory. Across 19 streams, an optional channel branch lifts injected single-sensor ER to 80.7%–96.5% while raising nominal causal FPR from 1.5% to 6.8%–19.4%; natural sensor failures and slow ramps ($L \geq 4C$) remain limitations.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep detectors typically forecast or reconstruct raw windows and threshold $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$: autoregressive long short-term memory (LSTM) forecasters [4] model spacecraft channels directly; TranAD [6] pairs adversarial training with self-conditioning attention; Anomaly Transformer [7] contrasts Gaussian prior and series-level attention; and TimesNet [5] reshapes 1D series along Fast Fourier Transform (FFT) periods for 2D convolution. Equal channel-wise reconstruction losses fit stochastic jitter, so summing errors across $K \geq 25$ channels yields heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) spikes and high false alarms under baseline shift [12].

Contrastive and Joint-Embedding Predictive Architectures: Representation methods encode windows into compact latents [13]. Contrastive approaches—temporal-contextual contrasting [14], TS2Vec [15], Neural Contextual Anomaly Detection (NCAD) [10] with synthetic anomalies, and dual-attention DCdetector [16]—depend on whether synthetic perturbations or patch discrepancies match test faults (0.088 and 0.097 stream Point-F1 for NCAD and DCdetector vs. 0.186 for latent prediction on 130 streams, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [17], [18] predict target embeddings z_{tgt} from context z_{ctx} without observation decoding or negative pairs, avoiding collapse via exponential moving average (EMA) targets [19] or variance-covariance terms [20]–[22].

Residual Decomposition, Conformal Thresholding, and Statistical Process Control: Telemetry reflects underlying system dynamics [23]. Neural ordinary differential equations (ODEs) [24], Physics-Informed Neural Networks (PINNs) [25], Hamiltonian/Lagrangian nets [26], [27], and Koopman lifts [28]–[32] model structured dynamics, though partial differential equation (PDE) penalties require known laws and exhibit stiff gradients on uncalibrated sensors [33]; we instead borrow the algebraic mean-plus-fluctuation identity of Reynolds averaging [34] strictly as a multi-horizon residual summary. Multivariate statistical process control (SPC) charts such as Hotelling's T^2 [35] and Multivariate Exponentially Weighted Moving Average (MEWMA) [36] collapse whitened coordinates into one scalar, as do 1D deep scores calibrated via extreme-value theory [37] and Generalized Pareto Distribution (GPD) tails in SPOT [8] under serial dependence [38]. Conversely, univariate Cumulative Sum (CUSUM) banks [39] and per-coordinate split-conformal rules under Bonferroni, Šidák, or Benjamini–Hochberg (BH) false-discovery-rate (FDR) control [40] hit a discrete calibration floor $(N_{\text{cal}} + 1)^{-1}$ on short histories, whereas matched-rate per-coordinate and recalibrated Mahalanobis conformal rules inflate under coordinate drift (Section VI-A). All methods are evaluated without point adjustment [12], [41], [42].

III. Multi-Sensor Telemetry Benchmark Suite

Our benchmark suite spans 45 continuous telemetry streams across 11 datasets from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [12], and mTSBench [45], expanded in archive-wide evaluations to 141 qualifying multivariate streams across 12 datasets (Section V-B). Thirty-three streams enter the development protocol (19 non-GHL streams and 14 GHL streams) and 12 are reserved with explicit accounting below. *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [46] (K=38), where NASA [4] labels each SMAP/MSL sequence as an out-of-range *point* fault or in-range *contextual* fault (Table IV(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [43] at 64 Hz (K=9, 72 episodes, where S=64 equals 1 s). *Process and infrastructure telemetry* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 slow-ramp faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry (K=9, 37 anomalies), Metro I-94 traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six

CICIDS network streams (K=72). Section VI-B evaluates six injected single-channel fault classes across all 19 multivariate development streams ($K \in [5, 55]$).

Each stream provides nominal history ($N_{\text{tr}} \in [2{,}074, 120{,}000]$) and labeled test data ($N_{\text{test}} \in [2{,}158, 114{,}001]$), standardized per channel on the weight-fitting prefix ($\boldsymbol{\sigma}$ floored at 10^{-2}) and split chronologically into training (65%, further split 50%/15% into weight-fitting and checkpoint-selection prefixes under disjoint selection; Section V-A), nominal calibration (17.5%), a 64-step isolation gap, and held-out nominal evaluation (17.5%). Development 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$), $K \leq 55$, and $N_{\text{test}} \leq 1.2\times10^5$ (Table III(c)); the 12 reserved streams comprise five short-nominal streams ($T_{\text{nom}} \in [764, 1{,}492]$, $N_{\text{cal}} \leq 101$: CalIt2, MSL C-2, MSL D-16, two Room-Occupancy), six CICIDS streams (K=72), and GECCO (K=9, included in archive cohorts). Archive cohorts evaluate all 141 qualifying streams across 12 datasets ($N_{\text{cal}} \geq 100$, ≥ 1 anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN; 118 untouched by design).

Two caveats bound external validity: SMAP and MSL contain documented labeling flaws [42], and streams within multi-stream archives share hardware platforms. Dataset clusters ($N \in \{7, 9, 12\}$) are therefore treated as co-primary units alongside streams via Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and LODO audits (Section V-A).

IV. Methodology: Multi-Horizon Laminar-JEPA and Distilled Laminar-JEPA-Lite

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Fig. 1 illustrates the four-stage pipeline from raw telemetry $\mathbf{X} \in \mathbb{R}^{T\times K}$ (K channels over T timesteps) to binary alarm decisions. A sliding window of length $W = C + S = 320$ advances one step at a time, splitting each position t into a historical *context window* $\mathbf{x}_{\text{ctx}}$ of $C = 256$ steps and a future *target horizon* $\mathbf{x}_{\text{tgt}}$ of $S = 64$ steps (ending at the causal decision timestamp $\tau = t + S - 1$). Before mixing channels, multichannel inputs ($K > 1$) pass through a causal depthwise 1D convolution ($K \leq 1$ uses an identity bypass) and are split across four smooth dyadic frequency shells on a 256-point real fast Fourier transform (rFFT) grid. The shells form an amplitude partition of unity within 1.2×10^{-7}, separating slow baseline drift from high-frequency vibration before per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$) and a softmax router feed a causal dilated temporal convolutional network (TCN) [48].

Rather than reconstructing raw sensor waveforms, a Joint-Embedding Predictive Architecture (JEPA) [9] maps context and target windows into compact $D = 32$-dimensional layer-normalized summaries via dual online (E_θ) and target (E_ϕ) encoders. Target weights track online weights via an exponential moving average (EMA) at momentum $m = 0.996$ (0.995 on archive cohorts) [17], [19], placing summaries near radius $\sqrt{D}$ up to the normalization epsilon. Deep dilated TCN stacks can smooth out brief onset spikes, while operating-regime shifts displace the latent mean. Both encoders address these two failure modes directly:

1) *Multi-depth hierarchy tapping:* Each encoder taps three intermediate feature maps ($d_{\text{model}} = 48$) from the shallow spectral stem, mid-depth TCN block, and final TCN block. Concatenating the channel-centered terminal state and temporal root-mean-square (RMS) energy of each tap, projecting to $\mathbb{R}^D$ through a two-layer GELU head, and blending with the deep TCN output through a bounded gate ($g \leq 0.5$) preserves fast transients alongside long-range context.
2) *Synchronized barycenter centering:* During training, an online centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ updates at rate 0.05, copies synchronously to E_ϕ, and is subtracted from both context and target summaries prior to final LayerNorm. Centering reduces learned common-mode offsets without guaranteeing invariance to unseen drift.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Variance-Floor Decorrelation Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor maps z_{ctx} to a single 64-step forecast $\hat{z}_{\text{tgt}}$, trained with predictive mean-squared error (MSE) and a VICReg-style log-determinant volume penalty ($\beta = 0.05$) [20]. Pooling an entire 64-step target window into one vector can average out short-lived faults, and LayerNorm bounds vector length without bounding temporal tail heaviness.

To track how prediction error evolves across the target window, Stage 2 of *Laminar-JEPA* encodes $M = 4$ nested target prefixes of lengths $S_m \in \{16, 32, 48, 64\}$ at $S = 64$ (clamped to at least two steps when $S > 1$; at $S = 1$, all four prefixes share the single target step and differ only through predictor rollout). Starting from z_{ctx} with learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, a recurrent sub-horizon predictor P_{sub} rolls out sequential LayerNorm forecasts $\hat{z}_{\text{tgt}}^{(1)}, \ldots, \hat{z}_{\text{tgt}}^{(M-1)}$ while an anchor head P_θ predicts the full horizon $\hat{z}_{\text{tgt}}^{(M)}$. Assigning half the loss weight to $m = M$ and splitting the other half equally across $m < M$, training minimizes multi-horizon MSE plus a non-collapse penalty $\boldsymbol{\Delta}_\gamma$:

$$\mathcal{L}_{\text{Lam}} = \sum_{m=1}^{M} \frac{w_m}{D}\left(\left\|\hat{z}_{\text{tgt}}^{(m)} - \text{sg}\left(z_{\text{tgt}}^{(m)}\right)\right\|_2^2 + \left\|\boldsymbol{\Delta}_\gamma\left(\hat{z}_{\text{tgt}}^{(m)}\right)\right\|_F^2\right), \tag{1}$$

where $\text{sg}(\cdot)$ stops gradients through E_ϕ. Unlike log-determinant penalties that require batch size $B > D$, $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ penalizes collapse directly on small batches: its diagonal entries $\sqrt{\max(0, \gamma - \sigma_d)}$ enforce a per-coordinate standard-deviation floor $\gamma = 0.75$ (using batch-biased variance plus 10^{-4}), and its off-diagonal entries $\Sigma_{dj}/\sqrt{2}$ penalize cross-coordinate sample covariance ($B - 1$ divisor). At $B = 16 < D = 32$, effective latent rank $\text{RankMe}(z_{\text{tgt}})$ rises from 11.6 to 19.8 [49].

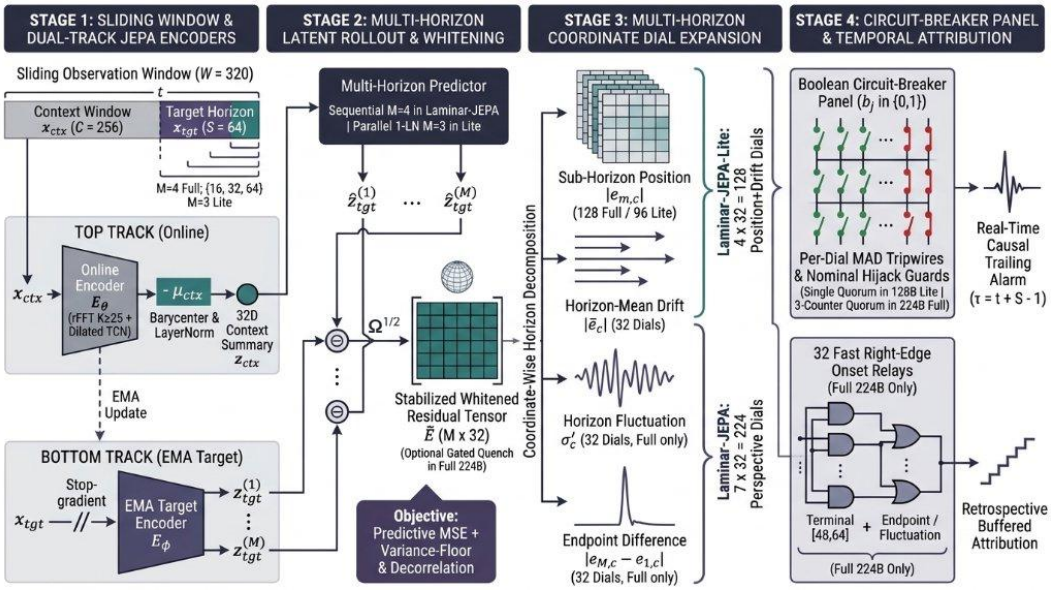


Fig. 1. Overview of the four-stage *Laminar-JEPA* and distilled *Laminar-JEPA-Lite* pipeline: **(1)** sliding window ($C{=}256, S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with conditional rFFT splitting ($K{\geq}25$ in Lite), barycenter centering, and LayerNorm; **(2)** multi-horizon rollout ($M{=}4$ sequential in full *Laminar-JEPA*; $M{=}3$ parallel single-LayerNorm in *Lite*) under variance-floor and decorrelation loss plus resolvent whitening into $\tilde{\mathbf{E}} \in \mathbb{R}^{M\times 32}$; **(3)** horizon expansion into $7 \times 32 = 224$ dials (full) or $4 \times 32 = 128$ position-plus-drift dials (*Lite*); and **(4)** per-dial Boolean circuit breakers emitting causal ($\tau{=}t{+}S{-}1$) or prefix-buffered alarms.

C. Stage 3: Calibration-Stabilized Whitening, Optional Weight-Space Quenching, and 224-Dial Expansion

Collapsing a multi-horizon residual tensor into a single scalar score creates a bottleneck: if nominal drift inflates even a few coordinates between training and calibration, those coordinates dominate the scalar norm. Stacking the $M = 4$ signed residuals $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ forms $\mathbf{E}(t) \in \mathbb{R}^{M\times D}$ ($MD = 128$ entries). On nominal training windows, eigendecomposing the residual covariance $\hat{\mathbf{\Sigma}}_{\text{res}} = \mathbf{U}\,\text{diag}(\lambda_k)\mathbf{U}^\top$ defines a damped resolvent precision matrix $\mathbf{\Omega} = \mathbf{U}\,\text{diag}\big(\lambda_k/(\lambda_k + 10^{-3})^2\big)\mathbf{U}^\top$. Multiplying centered residuals by $\mathbf{\Omega}^{1/2}$ yields whitened errors $e_{m,c}^{\text{white}}(t)$ while suppressing near-null eigenmodes instead of amplifying them. Stage 3 then insulates coordinates against nominal calibration drift and expands the horizon trajectory:

1) *Calibration-stabilized whitening and optional quenching:* If a coordinate's whitened RMS on nominal calibration windows exceeds 1.10, it blends smoothly toward its RMS-standardized raw residual $\hat{e}_{m,c}^{\text{raw}}(t)$ so no drifting eigenmode can blow up downstream tripwires:

$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t), \qquad w_{m,c} = \frac{1}{1 + 2\max\big(0, \text{RMS}_{\text{cal}}^{\text{white}}(m,c) - 1.10\big)^2}. \tag{2}$$

   Full *Laminar-JEPA* also includes an optional zero-gradient post-training *weight-space quench* fit strictly on nominal training and calibration windows. Coordinates are ranked by an instability score standardized via the median and median absolute deviation (MAD), combining train-to-calibration mean/scale shifts, calibration half-split drift, residual autocorrelation, and a negative guard for isolated nominal spikes. For candidate top-k drifting coordinates ($k \in \{4, 2\}$), four steps run sequentially: *(Q1)* damping the selected predictor output rows/LayerNorm gains and shifting $\mathbf{p}_m$ toward calibration means (which induces unselected spillover through 32D LayerNorm, audited in Section VI-A); *(Q2)* blending residual centers 75% toward post-edit calibration means; *(Q3)* attenuating off-diagonal precision couplings by 0.15 and rescaling whitened calibration RMS; and *(Q4)* retaining the edit (preferring $k = 2$ on ties) only if nominal half-split FPR improves while preserving at least 92% of pre-edit quorum thresholds, reverting otherwise.
2) *Multi-horizon expansion into $7D = 224$ dials:* Inspired algebraically by the mean-plus-fluctuation split of Reynolds averaging [34] across the $M = 4$ horizons of each coordinate $c \in \{1, \ldots, 32\}$, we separate $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into its horizon mean $\bar{e}_c(t) = M^{-1}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and zero-mean fluctuation $F_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$. Taking magnitudes yields $7D = 224$ nonnegative dials $\mathbf{d}(t) \in \mathbb{R}_{\geq 0}^{224}$: (a) $4D = 128$ *sub-horizon positions* $|\tilde{e}_{m,c}(t)|$; (b) $D = 32$ *horizon-mean drifts* $|\bar{e}_c(t)|$; (c) $D = 32$ *horizon fluctuations* $d_{\text{fluc},c}(t) = \sqrt{M^{-1}\sum_{m=1}^{M} F_{m,c}^2(t)}$; and (d) $D = 32$ *endpoint differences* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$.

Removing the horizon mean splits each coordinate's squared error orthogonally into persistent offset plus within-window fluctuation, and confines the $M \times M$ horizon Gram matrix to the $(M - 1)$-dimensional subspace orthogonal to $\mathbf{1}_M$ (a geometric property of the representation rather than an accuracy guarantee):

Proposition 1 (Horizon Energy Split and Spectral Concentration Bounds)**.** *For $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) with horizon mean $\bar{\mathbf{e}}(t)$ and fluctuation matrix $\mathbf{F}(t) = \tilde{\mathbf{E}}(t) - \mathbf{1}_M \bar{\mathbf{e}}(t)^\top \neq \mathbf{0}$: (i) for every coordinate c, $M^{-1}\sum_{m=1}^{M} |\tilde{e}_{m,c}(t)|^2 = |\bar{e}_c(t)|^2 + d_{\text{fluc},c}^2(t)$ and $d_{\text{accel},c}(t) \leq \sqrt{2M}\, d_{\text{fluc},c}(t)$; and (ii) the $M \times M$ horizon Gram matrix $\mathbf{G}(t) = \mathbf{F}(t)\mathbf{F}(t)^\top \in \mathcal{S}_+^M$ has ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ whose scale-invariant*

concentration ratio for any $p > 0$ [50] satisfies

$$R_{LT}\big(\mathbf{F}(t)\big) = \frac{\lambda_M^p}{\sum_{m=1}^M \lambda_m^p} \in \Big[\frac{1}{M-1},\, 1\Big], \tag{3}$$

reaching $\frac{1}{M-1}$ iff energy is isotropic across all $M-1$ active modes ($\lambda_2 = \cdots = \lambda_M > 0$, requiring $D \geq M-1$) and 1 iff $\mathbf{F}(t)$ has rank one. Empirically, scalar reweighting by R_{LT} has a null effect (Section VI).

Proof. Zero column sums ($\sum_{m=1}^M F_{m,c} = 0$) cancel the cross term in $M^{-1}\sum_{m=1}^M(\bar{e}_c + F_{m,c})^2$, and $(F_{M,c} - F_{1,c})^2 \leq 2(F_{M,c}^2 + F_{1,c}^2) \leq 2M\, d_{\text{fluc},c}^2$ gives (i). For (ii), $\mathbf{F}^\top \mathbf{1}_M = \mathbf{0}_D$ implies $\mathbf{G}\mathbf{1}_M = \mathbf{0}_M$ ($\lambda_1 = 0$), while $\|\mathbf{F}\|_F^2 > 0$ implies $\lambda_M > 0$. Summing $0 \leq \lambda_m^p \leq \lambda_M^p$ over $m \in \{2, \ldots, M\}$ yields $\lambda_M^p \leq \sum_{m=2}^M \lambda_m^p \leq (M-1)\lambda_M^p$; inverting and multiplying by λ_M^p gives (3) and its equality conditions, and nonzero scalar rescaling cancels. □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

Instead of adding all 224 dials into one global score, Stage 4 (Fig. 1, Algorithm 1) assigns each dial its own binary circuit breaker on nominal calibration data. Because the 224 dials are derived from a single 4×32 residual tensor, their trips are correlated (median effective rank 65 of 224 on the development cohort, with joint trip-count variance $7.2\times$ independent trials). Alarm quorums are therefore calibrated directly on the empirical distribution of simultaneous nominal trip counts:

1) *Per-dial standardization and hijack guards:* Each dial $j \in \{1, \ldots, 224\}$ is standardized against its nominal calibration median m_j and scaled MAD $s_j = 1.4826\ \mathrm{MAD}(d_{j,\text{cal}}) + 10^{-4}$ into a nonnegative score $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine coordinates that drift during calibration, its 98.5th-percentile tripwire τ_j is multiplied by two nominal hijack guards ($\gamma_j^{\text{cal}} \geq 1$ when second-half calibration 98th percentiles exceed the first half by $> 1.35\times$, and $\gamma_j^{\text{sc}} \geq 1$ when dial scale exceeds $1.5\times$ its perspective-group median by $> 1.25\times$):
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\, \gamma_j^{\text{cal}}\, \gamma_j^{\text{sc}}. \tag{4}$$
2) *32 right-edge onset relays:* When an abrupt fault first enters the right edge of the S-step target window, earlier prefixes ($m \in \{1, 2\}$) still cover nominal data. To catch these transitions promptly, each coordinate $c \in \{1, \ldots, 32\}$ defines stricter 99.3rd-percentile edge switches ($\gamma^{1.25}$ guard) and fires a binary onset relay $r_c(t) \in \{0, 1\}$ whenever a terminal-prefix trip ($m = 4$) coincides with an acceleration, fluctuation, or $m = 3$ trip on the same coordinate.
3) *Causal vs. buffered attribution and quorum voting:* Under *causal trailing attribution*, active breakers on window $[t, t+S)$ emit a real-time decision strictly at its right endpoint $\tau = t + S - 1$ (structural lag $S-1$, empirical delay $\Delta t = t_{\text{alm}} - t^*$). Under *retrospective buffered attribution*, position votes route to their corresponding prefixes $[t, t+S_m)$, drift and fluctuation votes span $[t, t+S)$, and endpoint-difference votes route to $[t+S_3, t+S)$ (empty at $S = 1$). Summing active trips yields $N_{224}(\tau)$, position subpanel count $N_{128}(\tau)$, and relay count $N_{\text{edge}}(\tau)$ (summed as $N_{\text{tot}} = N_{224} + N_{128} + 2N_{\text{edge}}$ for continuous PR-AUC), triggering a binary alarm whenever any counter exceeds its nominal quorum:
$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \ \vee\ N_{128}(\tau) > K_{128} \ \vee\ N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \tag{5}$$
with nominal calibration quorums $K_{224} = \max(1.12\, q_{0.995}, 5.0)$, $K_{128} = \max(1.12\, q_{0.995}, 3.5)$, and $K_{\text{edge}} = \max(1.25\, q_{0.999}, 2.5)$.

E. Autopsy-Distilled Architecture: Laminar-JEPA-Lite

Layer-by-layer activation tracing and component ablations of *Laminar-JEPA* (Section VI) motivate three simplifications that cut trainable parameters by 35%–52% (141.9k at $K < 25$; 222.3k at $K \geq 25$ vs. 295.8k) and accelerate windowed GPU inference by $2.43\times$ ($3.14\times$ at $K < 25$) with zero post-training weight editing:

1) *Four-block TCN and $K \geq 25$ spectral gating:* Because a fifth dilated block damps short-prefix anomaly divergence without improving effective rank, *Laminar-JEPA-Lite* uses four TCN blocks ($d \in \{1, 2, 4, 8\}$), 64D pooling heads, and activates the four-shell rFFT front-end only on high-channel streams ($K \geq 25$), using a direct causal projection when $K < 25$.
2) *Parallel three-horizon single-LayerNorm predictor:* Instead of sequential four-horizon unrolling and stacked predictor normalizations, a single residual two-layer GELU MLP predicts $M = 3$ nested prefixes ($S_m \in \{16, 32, 64\}$ at $S = 64$; 96×96 precision $\mathbf{\Omega}$) in parallel from $z_{\text{ctx}} + \mathbf{p}_m$ with one terminal LayerNorm, avoiding sequential lag and LayerNorm cross-coupling.
3) *128-dial position-plus-drift panel and single quorum:* Retaining only the 96 sub-horizon magnitudes $|\tilde{e}_{m,c}(t)|$ and 32 horizon-mean drift dials $|\bar{e}_c(t)|$ ($4D = 128$ breakers calibrated via (4)) removes fluctuation/endpoint dials, onset relays, and weight quenching, firing under one quorum $K_{\text{Lite}} = \max\big(1.12\, q_{0.995}(N_{128,\text{cal}}),\, 4.0\big)$.

1D Scalar Collapse Controls: To isolate circuit-breaker gains over scalar aggregation on identical backbones (Section VI), we evaluate 1D controls that pool a residual vector via an equal blend of its ℓ_2 norm and softmax-weighted coordinate RMS ($\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA applies this scalar pooler to $\mathbf{e}(t) \in \mathbb{R}^{32}$, whereas 1D Multi-Horizon JEPA combines the full-tensor and peak-horizon pooled scores in root-mean-square (with optional R_{LT} modulation $\alpha_{\text{LT}} \in \{0.12, 0\}$), calibrated via SPOT [8], [37], [51] ($u = p_{98}, q = 10^{-3}$).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures use context $C = 256$, target horizon $S = 64$ ($W = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), latent dimension $D = 32$, nominal training stride 16, and evaluation stride 1. Models are trained with AdamW [52] (learning rate 10^{-3}, weight decay 10^{-4}) and zero per-dataset tuning: batch 16 for 15–20 epochs on development cohorts, and batch 16–64 for 10–12 epochs with gradient clipping (1.0) and cosine annealing on archive cohorts.

Algorithm 1 Laminar-JEPA (224-Bit) and Laminar-JEPA-Lite (128-Bit)

Require: Nominal history $\mathbf{X}_{\text{nom}}$, test stream $\mathbf{X}_{\text{test}}$, tier $\in \{\text{Full}, \text{Lite}\}$, attribution mode $\in \{\text{causal}, \text{buffered}\}$.
Ensure: Per-timestep binary alarms $\hat{y}(\tau) \in \{0, 1\}$ and continuous score $N_{\text{tot}}(\tau)$.

Phase 1: Disjoint Training & Resolvent Whitening
Split $\mathbf{X}_{\text{nom}}$ chronologically into $(\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_{\text{cal}}, \text{gap}_S, \mathbf{X}_{\text{hold}})$ and standardize channels using $\mathbf{X}_{\text{fit}}$ alone.
Set nested prefixes $S_{1:4} = \{16, 32, 48, 64\}$ (Full) or $S_{1:3} = \{16, 32, 64\}$ with $K \geq 25$ rFFT gate (Lite).
for stride-16 training batches in $\mathbf{X}_{\text{fit}}$ **do**
    Encode $z_{\text{ctx}} = E_\theta(\mathbf{x}_{\text{ctx}})$ and targets $z_{\text{tgt}}^{(m)} = E_\phi(\mathbf{x}_{\text{tgt}}^{(m)})$ with barycenter shift $-\bar{\boldsymbol{\mu}}_{\text{ctx}}$.
    Roll out sequential $\{P_{\text{sub}}, P_\theta\}$ (Full) or parallel single-LayerNorm predictor (Lite); update θ via (1) and $\phi \leftarrow m\phi + (1-m)\theta$.
end for
Select checkpoint on $\mathbf{X}_{\text{sel}}$ (or fixed epochs) and fit $(\bar{\boldsymbol{\mu}}_{\text{res}}, \boldsymbol{\Omega} \in \mathcal{S}_+^{MD})$ on $\mathbf{X}_{\text{fit}}$.
if tier = Full and $q_{\text{wsp}} = 1$ **then**
    Rank drifting coordinates on $(\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{cal}})$; run Steps Q1–Q4 for $k \in \{4, 2\}$ and keep edit iff nominal split and 0.92-quorum gates pass.
end if

Phase 2: Nominal Circuit-Breaker Calibration on $\mathbf{X}_{\text{cal}}$
Extract stabilized whitened residuals $\tilde{\mathbf{E}}_{\text{cal}}$ via (2).
if tier = Full **then**
    Expand $\tilde{\mathbf{E}}_{\text{cal}}$ into $7D = 224$ dials (position, drift, fluctuation, endpoint) and 32 right-edge onset relays.
    Fit MAD scales (m_j, s_j), hijack guards γ_j, tripwires τ_j (4), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (5).
else ▷ Distilled Laminar-JEPA-Lite
    Extract $4D = 128$ position-plus-drift dials $(|\tilde{e}_{m,c}|, |\bar{e}_c|)$, fit τ_j (4), and set single quorum $K_{\text{Lite}} = \max(1.12\, q_{0.995}, 4.0)$.
end if

Phase 3: Online Inference on $\mathbf{X}_{\text{hold}}$ or $\mathbf{X}_{\text{test}}$
for each stride-1 window $[t - C, t + S)$ **do**
    Compute $\tilde{\mathbf{E}}(t) \to \mathbf{d}(t)$ and binary trips $b_j(t) = \mathbb{I}(z_j(t) > \tau_j)$ (plus relays $r_c(t)$ if Full).
    Attribute active counts to right edge $\tau = t + S - 1$ (causal) or prefix spans $[t, t + S_m)$ (buffered).
    Emit $\hat{y}(\tau) \in \{0, 1\}$ when active counts exceed $(K_{224}, K_{128}, K_{\text{edge}})$ (5) or K_{Lite}.
end for

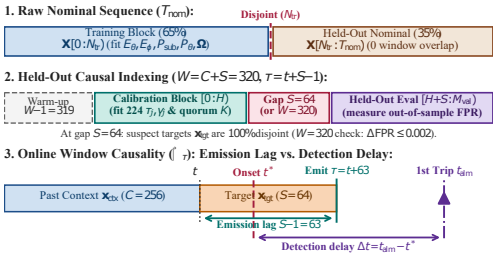


Fig. 2. Two-stage held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated nominal calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorums $(K_{224}, K_{128}, K_{\text{edge}})$, and (Row 3) structural emission lag ($S-1$) versus empirical fault-detection delay (Δt).

a) Disjoint Nominal Calibration Protocol, Boundary Isolation, and Cross-Fitting: Each stream's nominal history $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$): (1) *Stage A (Weight fitting vs. checkpoint selection):* Standardization statistics, encoder/predictor weights, running barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and precision $\boldsymbol{\Omega} \in \mathcal{S}_+^{MD}$ are fit on $\mathbf{X}_{\text{tr}}$. Because monitoring early-stopping loss on nominal calibration windows can make empirical calibration quantiles mildly optimistic, all archive benchmarks ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$, Table V) enforce a strictly *disjoint checkpoint-selection protocol* that splits $\mathbf{X}_{\text{tr}}$ chronologically into weight-fitting $\mathbf{X}_{\text{fit}}$ ($\approx 50\%$ of T_{nom}) and checkpoint-selection $\mathbf{X}_{\text{sel}}$ ($\max(W{+}64, \lfloor 0.15\, T_{\text{nom}} \rfloor)$ steps), leaving threshold calibration completely untouched by model training and selection (development cohorts train for fixed epochs on $\mathbf{X}_{\text{tr}}$ without validation early stopping). On short histories where carving out $\mathbf{X}_{\text{sel}}$ shrinks training support, a *two-fold chronological cross-fitting* check swaps $[0 : H)$ and $[H + S : M_{\text{val}})$ across two calibration folds; on the 19-stream cache, consensus voting ($\hat{y}^{(1)} \wedge \hat{y}^{(2)}$) lowers stream-mean FPR_{test} from 1.50% to 0.92% (median $1.07\% \to 0.38\%$) while retaining 66.7% median ER. (2) *Stage B (Circuit-breaker calibration and boundary isolation on $\mathbf{X}_{val}$):* Stride-1 windows of length $W = 320$ inside $\mathbf{X}_{\text{val}}$ yield $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window indices. Splitting at $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ and *held-out nominal evaluation block* $[H + S : M_{\text{val}})$, separated by an $S = 64$-step isolation gap $[H : H{+}S)$ so calibration and evaluation share zero target observations (a full $G = 320$ window gap changes causal FPR by < 0.25 percentage points). On $[0 : H)$ alone, the panel fits $w_{m,c}$ (2), dial medians m_j and MAD scales s_j, hijack guards γ_j, tripwires τ_j (4), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (5) (or single N_{128} quorum in *Laminar-JEPA-Lite*, τ_{SPOT} for 1D baselines), frozen on $[H + S : M_{\text{val}})$ and $\mathbf{X}_{\text{test}}$. We distinguish raw calibration window count $N_{\text{cal}} = H$ from autocorrelation-adjusted *effective sample size* $N_{\text{eff},s} = N_{\text{cal}}/\tau_{\text{int},s}$ using stream- and statistic-specific Sokal initial-positive-sequence estimators: across the 19 development streams, median τ_{int} is 2.51 (IQR $[1.52, 4.04]$) for individual dials z_j, 1.88 ($[1.22, 2.73]$) for N_{edge}, 5.58 ($[3.07, 14.09]$, max 206.0) for N_{224}, and 6.75 ($[5.10, 16.88]$) for 1D scalar scores (legacy lag-64 Bartlett pooled summary 2.66 median, 3.23 trimmed cohort summary; Section VII-E).

b) Hierarchical Statistical Units, Dataset-Clustered Inference, and Hypothesis Testing: Because streams within multi-stream archives share hardware platforms, we report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Dataset-clustered and stream-level inference:* Because stream-level p-values treat same-spacecraft channels as independent, *dataset-clustered inference* (95% cluster-bootstrap CIs and exact 2^D dataset sign-flip permutation tests) is emphasized alongside stream-level paired Wilcoxon tests and 95% stream-bootstrap CIs (2,000 resamples). For multiplicity reporting on the 118 untouched streams, the focal causal contrasts comparing *Laminar-JEPA-Lite* against Multi-Horizon JEPA (1D) and TS-JEPA yield Benjamini–Hochberg [40] adjusted stream-level $p = 0.0034$ on Point-F1 (raw $p = 0.0034$ and 1.6×10^{-4}) and adjusted dataset sign-flip $p = 0.039$ on ER (raw $p = 0.012$ and 0.039, positive on $8/9$ datasets); other quoted p-values are raw and descriptive. Non-Exathlon held-out subsets ($n = 101$, 8 datasets) verify insensitivity to archive composition. (2) *12-fold LODO audit:* Across all $12/12$ LODO folds on the 141-stream archive, *Laminar-JEPA-Lite* retains a positive causal Point-F1 gain over Multi-Horizon JEPA (1D) ($\Delta \in [+0.002, +0.039]$, $\max p = 0.015$) and TS-JEPA ($\Delta \in [+0.023, +0.034]$, $\max p = 3 \times 10^{-4}$), and selecting the detector maximizing

causal Point-F1 subject to $\text{FPR}_{\text{hold}} \leq 6\%$ on $D-1$ datasets chooses *Laminar-JEPA-Lite* in all $12/12$ folds.

B. Consolidated Cohort Organization, Development vs. Held-Out Separation, and Baseline Fairness

Across all evaluations, *Laminar-JEPA* denotes the sequential four-horizon 224-bit architecture with nominal split-gated quenching (*Laminar-JEPA (w/o Quench)* omits Steps Q1–Q4), and *Laminar-JEPA-Lite* denotes the parallel three-horizon 128-bit position-plus-drift architecture without weight editing. Both causal trailing and retrospective buffered attribution are paired on identical splits within each cohort (aside from a single-horizon 130-stream buffered baseline screen in Section VII-A): (1) *Development cohorts (design-inspected):* A recomputed 19-stream residual cache across 6 datasets evaluates the dial ladder, component ablations, and conformal/SPC comparators on identical 4H unquenched residuals (Table I(a,b), Table II); a separately trained 33-stream development benchmark across 7 datasets (19 non-GHL matched-backbone streams in Table I(c) plus 14 GHL streams in Table III) pairs *Laminar-JEPA*, *Laminar-JEPA-Lite*, 1D JEPAs, and observation-space baselines; and tripwire/quenching constants were fixed on 23 tuning streams (19 non-GHL plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, and `OPPORTUNITY S1-ADL3`). (2) *Untouched held-out cohorts (primary generalization evidence):* Excluding those 23 design streams from the 141-stream archive (12 datasets, strictly disjoint $\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$) leaves 118 **untouched held-out streams** across 9 datasets (101 non-Exathlon streams across 8 datasets; Table V), supplemented by a single-seed 4H Q1–Q4 step-isolation audit on the same 118 streams (Section VI-A). None of the 118 held-out streams was inspected during architecture or threshold design.

Standardized Baseline Fairness Protocol: Scalar baselines (TS-JEPA, Multi-Horizon JEPA (1D), TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], and DCdetector [16]) share the identical chronological splits, training-only standardization, isolation gaps, and SPOT calibration ($u = p_{98}$, $q = 10^{-3}$), whereas conformal and SPC detector-bank baselines in Table II share the identical multi-horizon whitened residuals and nominal calibration alert rate. *Laminar-JEPA-Lite* uses 141.9k trainable parameters at $K < 25$ and 222.3k at $K \geq 25$ versus 295.8k for *Laminar-JEPA* (206k–649k across external neural baselines; circuit-breaker panels add zero trainable weights). Runs used PyTorch 2.4 / CUDA 12.4 on an RTX 5090 (0.15 ms/window batched inference for *Laminar-JEPA-Lite* vs. 0.38 ms/window for *Laminar-JEPA*, $2.43\times$ speedup; 188 MB peak memory); code: https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics and Aggregation

To prevent point-adjustment inflation [41] and aggregation ambiguity, all binary false-alarm and Point-F1 metrics are computed *per valid timestep* τ (never point-adjusted and never per-window without timeline attribution): FPR_{hold} averages $\hat{y}(\tau)$ over valid timesteps in $[H+S : M_{\text{val}})$, FPR_{test} averages $\hat{y}(\tau)$ over nominal test timesteps ($y(\tau)=0$), and Point-F1, Precision, Recall, and PR-AUC (from $N_{\text{tot}}(\tau)$) score all valid test timesteps against $y(\tau) \in \{0,1\}$. Event-level ER measures the fraction of contiguous $y=1$ segments with ≥ 1 alarm, and delay $\Delta t = t_{\text{alm}} - t^*$ is the step lag from segment onset t^* to first in-segment alarm. Unless explicitly labeled *median* or *Dataset-Macro*, cohort headline numbers report *stream-level arithmetic means* of within-stream seed averages, paired with stream medians for skewed FPR distributions and non-deployable post-hoc test-FPR controls ($f^* \in \{0.01, 0.05\}$) strictly for diagnostic comparison.

VI. Ablation Studies and Analysis

A. Ablation 1: Perspective Expansion Ladder, Circuit-Breaker Components, and Weight-Space Quenching

Table I(a),(b) and Fig. 3(a) evaluate perspective expansion and panel components on the 19-stream residual cache (not paired to the separately trained run in Panel (c)):

1) *Operational trade-off between 160 and 224 dials:* While the 224-dial panel lifts causal Point-F1 from 0.013 to 0.071 over 1D scoring ($p = 0.0023$; buffered $+0.033$, $p = 0.53$), more dials are not automatically better: the 160-dial position-plus-drift subset (128 sub-horizons + 32 horizon means, single quorum, no relays) achieves higher causal Point-F1 (0.098), higher causal ER (59.1% vs. 56.5%), shorter causal delay (74.8 vs. 90.1 steps), and lower nominal FPR (0.55% held-out, 0.87% test vs. 0.76%, 1.50%) at 29% lower dial cost, whereas all 224 dials lead only in retrospective buffered Point-F1 (0.224 vs. 0.198). This directly motivates the $M=3$ position-plus-drift 128-dial panel in *Laminar-JEPA-Lite*. Scalar R_{LT} modulation remains null in Panel (c).
2) *Relays, single-quorum voting, and hijack guards:* Removing the 32 onset relays lowers causal Point-F1 slightly ($0.071 \rightarrow 0.068$), and replacing the three-counter rule (5) with a single 224-dial quorum lowers causal Point-F1 to 0.063 ($\text{FPR}_{\text{hold}} = 0.0055$; Table II). Conversely, removing hijack guards ($\gamma_j = 1$) raises causal Point-F1 to 0.073 and shortens delay to 70.1 steps on this cache, showing that drift guards can damp useful fault responses when calibration drift is mild. None removes the $S-1$ causal reporting lag.
3) *Quenching step isolation and LayerNorm coupling:* On single-seed 4H checkpoints across all 118 untouched streams (unquenched causal $\text{FPR}_{\text{hold}} = 6.76\%$, F1 $= 0.0901$, ER $= 63.8\%$), $k=4$ predictor row damping (Step Q1) shifts unselected coordinates by a median 11.1% of nominal RMS (45.5% shifting $> 10\%$) via 32D LayerNorm coupling without lowering FPR_{hold} (7.06%), while horizon-embedding shifts lower raw RMS on 68.2% of edited coordinates (4.1% spillover). Buffer recentering and precision decoupling alone (Steps Q2–Q3) drive the FPR drop ($\text{FPR}_{\text{hold}} = 6.71\%$, $\text{FPR}_{\text{test}} = 14.75\%$) at lower causal F1 (0.0826), whereas adding the nominal split gate (Step Q4, accepted on $39/118$ streams) reaches $\text{FPR}_{\text{hold}} = 6.62\%$ while preserving causal F1 (0.0904, ER $= 63.3\%$). Because

TABLE I
CIRCUIT-BREAKER DIAL EXPANSION, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

Perspective Expansion Stage	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
1D Scalar Collapse (1 Dial)	.013	.191	**.315**	.413	.459	**.550**	166.2	143.1
32-Bit Full Horizon (m=4)	.050	.135	.255	.381	.500	.243	123.4	65.5
128-Bit Sub-Horizons (m=1..4)	.063	.216	.294	**.449**	.556	.462	110.6	**38.8**
160-Bit (+32 Horizon Drift)	**.098**	.198	.304	.429	**.591**	**.550**	**74.8**	108.3
192-Bit (+32 Horizon Fluc.)	.067	.192	.299	.443	.573	.453	79.1	101.6
224-Bit (+32 Endpoint/Relays)	.071*	**.224**	.299	.448	.564	.506	90.1	83.7

* Versus 1D, causal F1 difference $+0.057$ (stream-bootstrap 95% CI $[0.016, 0.120]$; Wilcoxon $p = 0.0023$, 12/2 wins/losses). Buffered difference $+0.033$ (CI $[-0.017, 0.093]$, $p = 0.53$). The scalar spectral-modulation ablation in the separately trained study remains null.

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

Component / Quorum Variant	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
224-Bit Breaker (Full Panel)	.071	.224	.299	.448	**.565**	**.506**	90.1	83.7
224-Bit Strict (High Quorum K_{strict})	.047	.168	.299	.448	.494	.439	207.6	**54.3**
w/o 32 Right-Edge Relays	.068	.225	.299	**.452**	**.565**	**.506**	90.7	83.7
w/o Prefix Routing (Unif. Smear)	.071	.222	.299	.440	**.565**	.497	90.1	115.4
w/o Hijack Guards (γ_j=1)	**.073**	**.226**	**.299**	.448	**.565**	**.506**	**70.1**	83.3

Panels (a,b): 19 streams × 3 seeds on the same whitened-residual cache, without the stabilized blend in (2). Panel (c) uses a separate training run; cross-panel comparisons are not paired.

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall Auto.	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
Laminar-JEPA (224-Bit)	0.92	0.190	0.013	0.128	0.144	**0.244**	**0.311**	**0.407**	0.673	0.795	0.612	0.757
Laminar-JEPA (w/o Quench)	0.88	0.190	0.016	0.128	0.142	0.235	0.304	0.381	**0.695**	0.815	**0.631**	0.765
Laminar-JEPA-Lite (128-Bit)	2.33	0.182	0.010	0.061	0.133	0.242	0.301	0.397	0.650	0.811	0.560	**0.772**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Multi-Horizon JEPA (1D)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Multi-Horizon JEPA (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	0.235	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

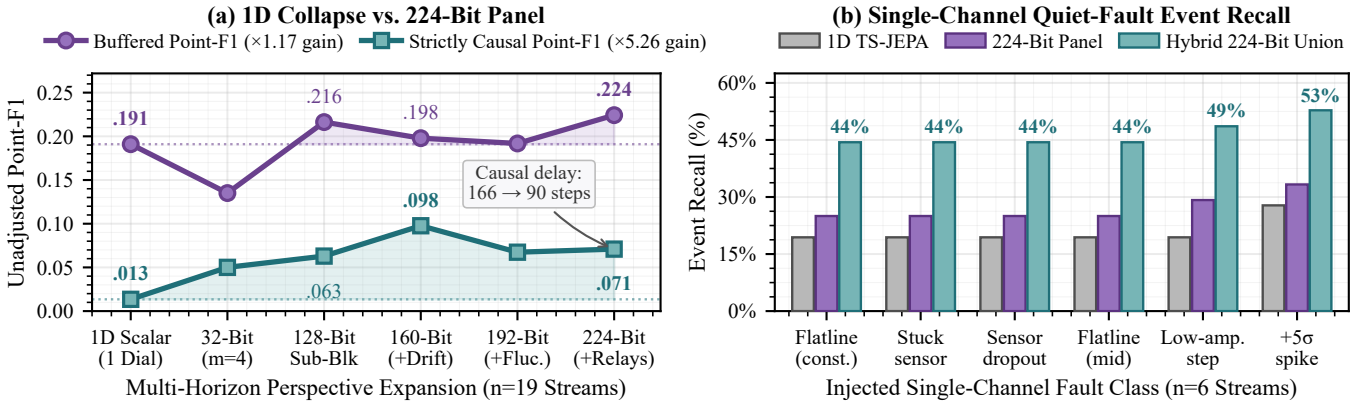


Fig. 3. Development studies. **(a)** Causal and buffered Point-F1 across the six-step expansion on cached whitened residuals from 19 streams (three seeds; no stabilization blend). **(b)** Event recall for injected single-channel faults in a separate six-stream study, comparing 1D TS-JEPA, 224-Bit Breaker, and their channel-level union.

ungated edits lower $k = 2$ causal F1 on $11/19$ development streams ($0.141 \rightarrow 0.049$) and $k \in \{8, 12\}$ cuts PR-AUC ($0.292 \rightarrow 0.223$), *Laminar-JEPA-Lite* omits weight editing altogether.

Conformal, multiplicity-controlled, and SPC detector banks on identical residuals: Table II benchmarks the panel against split-conformal, multiple-testing, and SPC detectors on the same 19-stream multi-horizon residuals without test-label tuning. Under family-wise Šidák/Bonferroni or Benjamini–Hochberg (0.5%) control across 128 coordinates or 224 dials, discrete calibration p-values have minimum resolution $(N_{\text{cal}} + 1)^{-1} > 3.9 \times 10^{-5}$, paralyzing causal recall (F1 ≤ 0.004). Matching per-coordinate conformal thresholds to the panel's nominal calibration alert rate (0.34%) causes tie-induced false-alarm inflation ($\text{FPR}_{\text{hold}} = 21.1\%$), as do recalibrated Mahalanobis conformal, Hotelling T^2, MEWMA, and CUSUM charts (27.9%–58.6%). By contrast, training-whitened multivariate ℓ_2 split-conformal scoring (F1 $= 0.072$, ER $= 60.8\%$, $\text{FPR}_{\text{hold}} = 0.62\%$), Fisher/Stouffer p-value pooling ($0.109/0.095$), and the 160-dial subset (0.098) match or exceed the 224-dial panel on this cache, demonstrating that training-fixed whitening and bounded coordinate aggregation—not 224-dial complexity alone—drive false-alarm stability.

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

Even when flatlining shifts the normalized summary (0.64 vs. 0.77 relative divergence for a variance-matched harmonic flip on `SMAP P-3`), terminal LayerNorm spreads static error thinly across all 32 coordinates (12.5 vs. 62.1 switches; 6.3% vs. 97.7% window alerts), and single-sensor faults suffer further $O(1/K)$ dilution across $K - 1$ nominal channels. Extending Fig. 3(b) (n=6), we evaluate a causal channel branch across all 19 multivariate development streams

TABLE II
CONFORMAL AND SPC COMPARATORS ON CACHED RESIDUALS

Detector (Calibration Rule)	Pt-F1 ↑	ER ↑	$\mathrm{FPR}_{\mathrm{hold}}$ ↓	$\mathrm{FPR}_{\mathrm{test}}$ ↓
224-dial panel (deployed)	0.071	0.565	0.007,6	0.015,0
160-dial P+D (single quorum)	0.098	0.591	0.005,5	0.008,7
224-dial single quorum (no relay)	0.063	0.564	0.005,5	0.008,8
Per-coord. conformal (Šidák .005)	0.000	0.000	0.000,0	0.000,0
224-dial BH FDR (q=.005)	0.004	0.018	0.000,0	0.000,1
Per-coord. conformal (matched)	0.265	0.843	0.211,1	0.283,1
Multivar. conf. ($\|\mathbf{e}_{\mathrm{white}}\|_2^2$, matched)	0.072	0.608	0.006,2	0.009,5
Multivar. conf. (Mahal., α=.005)	0.279	0.896	0.278,9	0.473,9
Fisher p-pool (matched)	0.109	0.591	0.010,5	0.018,3
Maximum dial z (matched)	0.083	0.565	0.009,3	0.015,7
Hotelling T^2 (matched)	0.283	0.896	0.290,1	0.483,1

19 streams × 3 seeds, causal means. Matched rules use each stream's nominal panel alarm rate (mean 0.34%; Stouffer F1 .095, $\mathrm{FPR}_{\mathrm{hold}}$ .0087; Top-k F1 .070, $\mathrm{FPR}_{\mathrm{hold}}$ .0077; MEWMA .4301, CUSUM .5862 $\mathrm{FPR}_{\mathrm{hold}}$).

($K \in [5, 55]$) without per-stream or test-label tuning. Per channel, three 64-step rolling statistics—log-variance drop $-\ln(\mathrm{Var}_{64}(x_c) + 10^{-4})$, training-fit first-order autoregressive (AR(1)) residual RMS, and mean shift $|\bar{x}_{64,c} - \mu_c|/\sigma_c$—are thresholded at $1.25\, q_{0.999}$ on nominal validation windows (floor 6.0 MAD, 1.25× half-split drift multiplier) with a 24-of-40-step persistence gate. On the highest-variance active channel $c^\star = \arg\max_{c:\sigma_c^{\mathrm{tr}} > 10^{-8}} \sigma_c^{\mathrm{tr}}$, a deterministic 9-point grid over $[320, N_{\mathrm{test}} - 520]$ (stride-50 fallback) places three nonoverlapping $L_{\mathrm{inj}} = 200$-step faults (57 events/class, gap ≥ 264) inside clean segments ($y^{\mathrm{orig}} \equiv 0$ on $[s - 64, s + 264)$). For $s \leq t < s + 200$ using training $(\mu_c, \sigma_c, q_{.995,c})$:

$$x'_{t,c^\star} = \begin{cases} \mu_{c^\star} & \text{variance collapse,} \\ \mu_{c^\star} + 1.35(-1)^{t-s}\sigma_{c^\star} & \text{prediction shock,} \\ x_{s-1,c^\star} & \text{sensor freezing,} \\ \max(q_{.995,c^\star}, \mu_{c^\star} + 2.2\sigma_{c^\star}) & \text{saturation,} \\ x_{t,c^\star} + 2.5\sigma_{c^\star}(t-s)/199 & \text{gradual degradation,} \\ x_{t,c^\star} + 1.5\sigma_{c^\star} & \text{isolated bias.} \end{cases} \tag{6}$$

Outside $[s, s + 200)$, all channels remain unchanged. Injected-event recall rises from 63.2% (standalone panel) to 93.0% for variance collapse, 63.2% to 94.7% for freezing, and 82.5% to 94.7% (96.5% full branch) for saturation under the variance-collapse union; the full channel branch raises recall from 75.4% to 96.5% on prediction shocks, 59.6% to 80.7% on gradual degradation, and 68.4% to 86.0% on isolated bias. However, on the *unmodified* test streams, causal $\mathrm{FPR}_{\mathrm{test}}$ rises from 1.5% (standalone panel, F1 = 0.071) to 6.8% (variance-collapse union, F1 = 0.070) and 19.4% (full branch, F1 = 0.129). Synthetic single-channel injections do not establish detection on natural sensor failures, and the full union's 19.4% false-alarm rate makes it unsuitable as a default detector.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining the multi-horizon JEPA backbone with and without 4-shell rFFT splitting on five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, `GECCO`; three seeds) reveals a clear channel-count threshold: shell splitting improves high-channel spacecraft detection (K=55 `MSL M-1` buffered F1 0.085 → 0.410; K=25 `SMAP P-3` causal F1 0.212 → 0.321) by separating thermal drift from actuator chatter, but lowers F1 at K=9 (`Daphnet`, `GECCO`), setting the $K \geq 25$ spectral gate in *Laminar-JEPA-Lite*.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 16 matched-budget arms in Table I(c) isolates five mechanisms: (1) raw-observation forecasting and autoencoding share heavy tails ($\gamma_2 \approx 6.2$), large drift ($D_{\mathrm{KS}} \approx 0.45$), and ≈ 25% causal FPR; (2) a D=32 LayerNorm bottleneck in observation space cuts kurtosis ($\gamma_2 = 0.91$) and halves FPR, yet raw targets keep drift high ($D_{\mathrm{KS}} = 0.376$); (3) learned D=32 latent targets and whitening halve drift ($D_{\mathrm{KS}} = 0.214$ in TS-JEPA and 0.179–0.190 across multi-horizon JEPA tiers, $p \leq 0.001$); (4) linear PCA and Isolation Forest keep low FPR but miss nonlinear faults; and (5) frozen targets and ℓ_2 scoring retain $< 2\%$ causal FPR, whereas latent context autoencoding lowers causal F1.

VII. RESULTS AND DISCUSSION

A. Development 33-Stream and Untouched Held-Out 118/141-Stream Benchmark Results

Following the cohort hierarchy in Section V-B, Table III reports the 33-stream development benchmark under retrospective buffered attribution (Panel (a)), causal streaming (Panel (b)), and per-dataset stratification (Panel (c)), while Table V reports the strictly disjoint-selection ($\mathbf{X}_{\mathrm{sel}} \perp \mathbf{X}_{\mathrm{cal}}$) benchmark across all 118 untouched held-out streams and the 141-stream archive. On a separate 130-stream baseline screen, TS-JEPA reaches stream-mean Point-F1 0.186 versus TranAD 0.152, TimesNet 0.151, DCdetector 0.097, NCAD 0.088, and Anomaly Transformer 0.082. The observed reduction in nominal false alarms with latent scoring is specific to these protocols; spherical normalization alone does not prove a distributional guarantee.

Operational trade-offs and configuration guidance: On the 33-stream benchmark, *Laminar-JEPA* edges out *Laminar-JEPA-Lite* on unweighted buffered Point-F1 (0.244 vs. 0.242) and causal ER (67.3% vs. 65.0%), whereas *Laminar-JEPA-Lite* halves buffered $\mathrm{FPR}_{\mathrm{hold}}$ (6.1% vs. 12.8%), leads 7-dataset macro Point-F1 (0.171 vs. 0.133), and cuts parameters by 35%–52% and inference latency by 2.43×. In practice, *Laminar-JEPA-Lite* is the recommended default for streaming and buffered deployment, *Laminar-JEPA* suits settings prioritizing peak buffered F1 or nominal 4H drift damping, short horizons ($S \leq 4$) address sub-window reporting lag, and raw-level variance or trend charts should be paired only when targeting single-sensor freezes or slow monotonic process ramps.

Scope boundaries and controlled slow-ramp sweep: Observation-space reconstructors show high nominal false-alarm rates here, whereas learned latent targets lower false alarms without guaranteeing light tails on arbitrary data. Conversely, because the encoder summarizes only the preceding $C = 256$ steps, gradual drift that shifts negligibly across one context window causes context and target summaries to co-move, explaining the 0.000 Point-F1 on the 14 GHL slow-ramp streams (low false alarms without recall is not detection). To map this boundary, we injected linear ramps of amplitude $a \in \{0.5, 1, 2, 4\}\sigma$ and duration $L \in \{64, 256, 1024, 4096\}$ (slopes a/L from 6.25×10^{-2}

TABLE III
PRIMARY 33-STREAM DEVELOPMENT BENCHMARK ACROSS ATTRIBUTION REGIMES

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-horizon coordinate circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.128	**0.244**	**0.407**	**0.669**	432.9	0.013	0.144	**0.673**	249.9	0.004	0.000	0.133
Laminar-JEPA-Lite (128-Bit)	0.061	0.242	0.397	0.636	59.3	0.010	0.133	0.650	73.8	0.005	0.000	**0.171**
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Multi-Horizon JEPA (1D)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Multi-Horizon JEPA (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	F1_{all}	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.214	0.179	0.145	**0.229**	0.144	**.009/.061**	.012/.091	.030/.091	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.003/.004**	.005/.005	.004/.004	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.078	**0.191**	0.000	0.000	0.000	.009/.104	.008/.020	.004/.017	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.284**	0.210	0.161	0.158	0.151	.013/.151	**.004/.049**	.054/.095	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.357	0.375	**0.379**	0.289	0.167	.021/.223	.016/**.066**	**.014**/.135	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	.010/.014	.001/**.000**	.008/.014	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.000	**0.240**	0.119	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), trained separately from Table I(a,b); best in **bold**, second underlined. In (c), n = streams, K = channels, E = events, π = anomaly rate, $\text{F1}_{\text{all}} = 2\pi/(1+\pi)$, Lam. = Laminar-JEPA, Lite = Laminar-JEPA-Lite. Unweighted Laminar-JEPA Δt reflects one late SWAN trigger (0.0002 recall; median Δt is 50.5 buffered / 45.8 causal, or 54.8 / 57.2 ex-SWAN). 95% stream-bootstrap CIs (n=19): Lite buffered FPR [.030, .103], F1 [.151, .335] and causal FPR [.006, .014], F1 [.055, .236]; Lam. buffered F1 [.150, .357] and causal FPR [.009, .018], F1 [.058, .257] (vs. Multi-Horizon 1D [.013, .097] and TimesNet causal FPR [.123, .375]).

to $1.22\times10^{-4}\sigma$/step, plus a 256-step plateau; 15 events/cell across 10 clean test segments on $\lceil 0.3K \rceil$-channel and 1-channel subsets). For steep multi-channel ramps ($L = 64$, $a \in [1, 4]\sigma$), the 224-bit panel reaches 86.7%–88.7% causal detection probability at 46–54-step median delay and 1.7%–1.9% false-alarm rate (+5.7–7.7 percentage points above the clean-segment Poisson episode baseline; +19.4–24.1 points for 1D SPOT at 60–67-step delay). By $L = 256$ ($= C$), panel excess causal detection narrows to +0.5–2.5 points (50–71-step delay), and for $L \geq 1024$ (slope $\leq 3.9\times10^{-3}\sigma$/step) raw hit rates (87%–91%) reflect background episodes over 1,280–4,352-step spans, with chance-corrected detection falling to $[-4.8, +0.8]$ points across all JEPA detectors ($[-5.9, -1.0]$ on single-channel ramps). Fixed-window JEPAs thus target abrupt to medium-onset faults rather than gradual industrial drift.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table III shows why 1D latent detectors degrade under real-time causal streaming. Buffered attribution averages residuals across overlapping predictions covering each timestamp ($\gamma_2 = 1.25$), whereas causal streaming evaluates a single right-edge step without temporal averaging ($\gamma_2 = 0.92$–2.33). Summing whitened coordinates into a 1D score allows one or two drifting coordinates to inflate the global SPOT threshold onto that heavier tail, missing sustained faults. Replacing the scalar sum with independently calibrated coordinate breakers isolates drifting coordinates. Against a rate-matched Bernoulli alerter $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, random guessing catches only 28.1% of events at our $\leq 1.3\%$ causal FPR, whereas at TimesNet's 24.3% causal FPR, coin flipping catches 60.0% of events by chance alone, exceeding TimesNet's 58.8% recall.

TABLE IV
CONTEXTUAL VS. POINT ANOMALY DETECTION

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**ER**	**FPR**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	0.519	0.775	0.022	0.800	**0.960**	0.783	0.900
Laminar-JEPA-Lite	0.532	**0.875**	0.013	**0.840**	0.920	0.700	0.967
Multi-Horiz. 1D	**0.652**	0.750	**0.006**	0.820	0.920	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=.05	f^*=.01	f^*=.05	q^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	**0.669**	0.448	**0.650**	0.835	0.644	0.795
Laminar-JEPA-Lite	0.655	0.376	0.567	**0.848**	0.581	**0.801**
Multi-Horiz. 1D	0.628	0.498	0.590	0.782	0.683	0.781
TS-JEPA	0.614	**0.520**	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Event recall (ER; autonomous vs. diagnostic f^*); best in **bold**, second underlined. (a) 10 SMAP/MSL streams by NASA taxonomy [4] (contextual: E-1, A-2, A-7, M-1, C-1 [2100, 2210]; point: A-1, P-3, D-3, D-14, D-15, C-1 [550, 750]). (b) n=19 non-GHL streams split into in-range ($\max|z| \leq 5\sigma$) vs. out-of-range ($\max|z| > 5\sigma$) causal events.

C. Multi-Horizon Scaling Across $S \in \{1, 4, 16, 32, 64\}$ and Latency Trade-Offs

Because a target window of length S emits its causal decision at $\tau = t + S - 1$, $S = 64$ imposes a 63-step reporting lag. We retrain on five selected streams at $S \in \{1, 4, 16, 32, 64\}$ to examine the latency trade-off. At $S = 1$, all predictor states are compared with the same one-step target, so this setting has no distinct target-horizon decomposition; at $S = 4$, the shortest encoded prefix is two steps. In the buffered $S = 1$ setting, endpoint-difference votes have no routed span. These short-horizon results therefore test the implemented detectors, not distinct horizon views.

In Fig. 4(a), both *Laminar-JEPA* and *Laminar-JEPA-Lite* exceed 1D Multi-Horizon SPOT across all five horizons while holding test FPR below 3.6% ($\leq 2.4\%$ for *Laminar-JEPA-Lite*, held-out FPR $\leq 1.2\%$, vs. $2.5\% \rightarrow 21.3\%$ for TimesNet). At

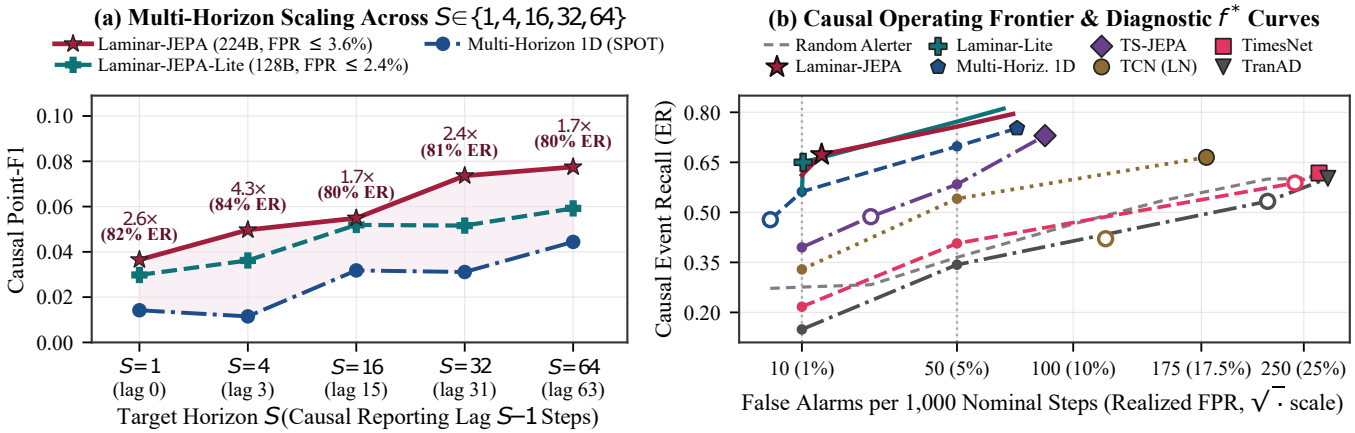


Fig. 4. Multi-horizon scaling ($S \in \{1, 4, 16, 32, 64\}$, causal reporting lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps, n=5 streams) and causal operating frontier (n=19 streams). **(a)** Causal Point-F1 and event recall (ER) across S comparing *Laminar-JEPA (224B)*, *Laminar-JEPA-Lite (128B)*, and Multi-Horizon 1D SPOT (nominal FPR $\leq 3.6\%$ across all S, vs. $2.5\% \rightarrow 21.3\%$ for TimesNet). **(b)** Causal ER vs. realized false alarms ($\sqrt{\cdot}$ scale) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted guides), autonomous points (*Laminar-JEPA* star, *Laminar-JEPA-Lite* cross, 1D open circles), and quantiles $q^* \leq 0.05$.

$S = 1$ and $S = 4$, *Laminar-JEPA* reaches $2.6\times$ and $4.3\times$ the scalar causal Point-F1 (0.037 and 0.050 vs. 0.014 and 0.012) with 77%–84% ER, while *Laminar-JEPA-Lite* reaches $2.1\times$ and $3.1\times$ (0.030 and 0.036) with 64%–82% ER; at $S = 4$, detection delay drops to 36.7–44.1 steps. Buffered Point-F1 peaks at 0.290 for *Laminar-JEPA* ($S = 32$) and 0.255 for *Laminar-JEPA-Lite* ($S = 64$) versus 0.182 for 1D SPOT. Five selected streams cannot establish a deployment rule across sampling rates or datasets.

D. Operating Budgets and Contextual vs. Point Anomalies

Fig. 4(b) and Table IV examine detection across operating budgets and fault taxonomies:

1) *Autonomous vs. diagnostic operating budgets (Fig. 4(b), Table I(c)):* Along the causal recall–FPR frontier, the autonomous points of *Laminar-JEPA* and *Laminar-JEPA-Lite* occupy the $\leq$ 1.3% held-out FPR regime without test-label tuning. At post-hoc diagnostic test-FPR controls ($f^* \in \{1\%, 5\%\}$), *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% (63.1% unquenched) and 56.0% of events at $f^* = 1\%$ versus 39.5% for TS-JEPA, 21.7% for TimesNet, and 14.9% for TranAD.
2) *In-range contextual vs. point faults (Table IV):* Under NASA's spacecraft taxonomy [4] (Panel (a)) and a 5σ amplitude split across all 19 development streams (Panel (b)), out-of-range point spikes ($> 5\sigma$) are flagged by reconstructors and amplitude rules alike, whereas in-range contextual faults decouple cross-channel phase or velocity inside normal amplitude bounds: on NASA contextual sequences, *Laminar-JEPA-Lite* and *Laminar-JEPA* achieve 87.5% and 77.5% autonomous ER at 1.3% and 2.2% FPR (84.0% and 80.0% at $f^* = 1\%$; 0.532 and 0.519 buffered PR-AUC) versus 62.5% at 41.5% FPR (56.0% at $f^* = 1\%$, 0.317 PR-AUC) for TimesNet.

E. Untouched Held-Out Generalization, Calibration Sample-Size Reliability, and Drift Decomposition

Table V and archive-wide audits establish three generalization and calibration findings:

TABLE V
DISJOINT-SELECTION HELD-OUT ARCHIVE BENCHMARK

	Pt-F1 ↑		PR-AUC ↑		ER ↑		FPR_{hold} ↓	
(a) Untouched n=118	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
TS-JEPA (1D)	.060	.131	.153	.200	.423	.475	**.042**	**.109**
Multi-Horizon JEPA (1D)	.074	.138	**.169**	**.234**	.495	.552	.059	.178
Laminar-JEPA (w/o Quench)	.077	.146	.147	.229	.639	.626	.056	.181
Laminar-JEPA (224B)	.073	.141	.144	.222	.621	.613	.054[§]	.174
Laminar-JEPA-Lite (128B)	**.085**[†]	**.153**[‡]	.148	.220	**.670**[¶]	**.686**	.057	.120
All Archive n=141 (12 Datasets)								
TS-JEPA (1D)	.055	.136	.165	.213	.426	.475	**.036**	**.103**
Multi-Horizon JEPA (1D)	.070	.146	**.182**	**.246**	.492	.543	.050	.163
Laminar-JEPA (224B)	.069	.150	.159	.237	.634	.623	.046	.161
Laminar-JEPA-Lite (128B)	**.084**	**.155**	.161	.233	**.665**	**.679**	.050	.107

	Causal Pt-F1 ↑			Buffered Pt-F1 ↑			Causal FPR_{hold} ↓		
(b) Held-out	n	**Lite**	**Lam.**	**1D**	**Lite**	**Lam.**	**1D**	**Lite**	**Lam.**	**1D**
SMAP	40	.097	.077	**.141**	.177	.175	**.201**	**.048**	.062	.125
SMD	18	.052	**.074**	.027	.124	**.135**	.067	.016	.031	**.004**
Exathlon	17	**.178**	.138	.085	**.217**	.168	.131	.181	.113	**.098**
Daphnet	15	**.088**	.070	.037	**.150**	.138	.093	.081	.075	**.009**
MSL	13	.023	**.032**	.026	.104	.112	**.211**	.010	.013	**.004**
OPPORT.	12	**.029**	.016	.010	**.093**	.048	.043	.011	.008	**.004**

Three seeds under $\mathbf{X}_{sel} \perp \mathbf{X}_{cal}$; best in **bold**, second underlined (1D = Multi-Horizon JEPA 1D). [†]Lite vs. 1D (n=118): causal F1 Δ=+0.012 (69/39 wins, p=0.0034; vs. TS-JEPA +0.025, cluster CI [+0.005, +0.046], $p = 1.6\times10^{-4}$); positive in 12/12 LODO folds on n=141 (max p=0.015). [‡]Buf. F1 Δ=+0.015 (p=0.0067; vs. Lam. +0.012, p=0.026, median buf. FPR_{hold} 1.96% vs. 4.74%). [¶]Causal ER Δ=+0.175 vs. 1D (cluster CI [+0.105, +0.273], 8/9 sets, sign-flip p=0.012; Lam. +0.126, p=0.016). [§]Quenching vs. unq. 4H: causal FPR_{hold} $\Delta = -0.002$ (7/9 sets improved, 0 worse, sign-flip p=0.016, Wilcoxon $p = 2.7\times10^{-4}$; median 0.96% vs. 1.25%).

1) *Independent generalization under disjoint checkpoint selection (Table V):* Across the 118 untouched streams ($\mathbf{X}_{sel} \perp \mathbf{X}_{cal}$, three seeds), *Laminar-JEPA-Lite* raises causal Point-F1 from 0.074 (Multi-Horizon 1D) and 0.060 (TS-JEPA) to 0.085 (69/39 wins, $p = 0.0034$; +0.025 vs. TS-JEPA, cluster CI $[+0.005, +0.046]$, $p = 1.6 \times 10^{-4}$). It lifts buffered Point-F1 from 0.138 to 0.153 ($p = 0.0067$; vs. 0.141 for *Laminar-JEPA*, $p = 0.026$, cutting median buffered FPR_{hold} from 4.74% to 1.96%), raises causal ER from 49.5% to 67.0% across 8/9 datasets (cluster CI $[+10.5\%, +27.3\%]$, sign-flip $p = 0.012$; *Laminar-JEPA* 62.1%, $p = 0.016$), and halves causal delay (62.7 vs. 141.9 steps). Excluding Exathlon ($n = 101$, 8 datasets), *Laminar-JEPA-Lite* holds causal FPR_{hold} at 3.7% (median 0.84%) and lifts causal ER from 47.7% to 63.5%; across all 141 streams (12 datasets), it stays positive in 12/12 LODO folds against Multi-Horizon 1D ($\max p = 0.015$) and TS-JEPA ($\max p = 3 \times 10^{-4}$). Nominal split-gated quenching in *Laminar-JEPA* lowers 4H causal FPR_{hold}

across $7/9$ datasets with zero worsened ($5.57\% \to 5.37\%$, median $1.25\% \to 0.96\%$, sign-flip $p = 0.016$, Wilcoxon $p = 2.7 \times 10^{-4}$) while moving causal Point-F1 from 0.077 to 0.073.

2) *Autocorrelation and calibration sample size:* Across the 19 development streams, Sokal autocorrelation τ_{int} spans median 2.51 (IQR $[1.52, 4.04]$, max 63.5) for dials z_j, 1.88 ($[1.22, 2.73]$, max 79.0) for N_{edge}, 5.58 ($[3.07, 14.09]$, max 206.0) for N_{224}, and 6.75 ($[5.10, 16.88]$) for 1D scores ($\tau_{\text{int}} \approx 3.23$ is only a trimmed summary). At median $N_{\text{cal}} = 408$, median N_{eff} (expected tail count) is 154.6 (2.32) at $q_{0.985}$ for z_j, 67.1 (0.34) at $q_{0.995}$ for N_{224}, and 195.9 (0.20) at $q_{0.999}$ for N_{edge}; even at $N_{\text{eff}} = 300$, expected tail counts $(4.5, 1.5, 0.45, 0.3)$ do not establish precise extreme-quantile estimation. Moving-block bootstrap ($L_{\text{blk}} \in \{1..128\}$) relative 95% CI half-widths span 12.7%–17.5% on z_j and 19.5%–25.0% on $(N_{224}, N_{\text{edge}})$, where the $1.12\times$ margin and floors $(5.0, 2.5)$ hold median FPR_{hold} to 0.36% and 0.00%. On the 118 untouched streams, $N_{\text{cal}} < 250$ ($n = 12$) averages 16.1% causal FPR_{hold} for *Laminar-JEPA-Lite* (17.7% *Laminar-JEPA*) versus 1.35% (2.56%) for $N_{\text{cal}} \geq 2{,}500$ ($n = 22$); Harrell–Davis Beta quantiles (1.26% vs. 1.67% exceedance at $N_{\text{cal}} = 250$) and IQR extrapolation remain exploratory.

3) *Regime drift vs. tail misfit via permutation decomposition:* Refitting thresholds across chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and pointwise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) splits is consistent with inter-block regime drift without ruling out parametric tail misfit: $\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$ represents 71.3% of excess causal FPR in TS-JEPA and 78.3%–81.8% in reconstructors, versus 18.7% in the centered model. Within-checkpoint traces of *Laminar-JEPA* (n=6, $K \in \{9, 25, 55\}$) show pre-tap vectors clustering on an off-center cap ($\|\bar{\boldsymbol{\mu}}_{\text{ctx}}\|_2 \in [4.73, 5.18]$ vs. $\sqrt{32} \approx 5.66$) at 0.06–0.16 anomaly divergence, whereas tap fusion and barycenter subtraction restore divergence to 0.70–0.94 and cut mean coordinate D_{KS} from 0.140 to 0.096.

VIII. Conclusion and Future Work

We evaluated *Laminar-JEPA* and its autopsy-distilled counterpart *Laminar-JEPA-Lite* across 33 development and 118 untouched held-out streams:

1) *Flagship vs. distilled circuit-breaker panels:* On the 33-stream benchmark, *Laminar-JEPA* and *Laminar-JEPA-Lite* reach 0.244 and 0.242 buffered Point-F1 and 67.3% and 65.0% causal ER at 1.3% and 1.0% FPR_{hold}, while *Laminar-JEPA-Lite* cuts parameters by 35%–52%, speeds up inference by $2.43\times$, and leads 7-dataset macro Point-F1 (0.171). On the 19-stream cache, the 224-dial panel (0.071, $p = 0.0023$), 160-dial subset (0.098), and training-whitened ℓ_2 conformal rule (0.072) outperform 1D collapse (0.013). Across 118 untouched streams ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$), *Laminar-JEPA-Lite* lifts causal Point-F1 from 0.074 to 0.085 ($p = 0.0034$) and causal ER from 49.5% to 67.0% across $8/9$ datasets ($p = 0.012$; $12/12$ LODO folds on $n = 141$), while nominal quenching lowers 4H causal FPR_{hold} across $7/9$ datasets ($p = 0.016$).

2) *Mechanisms and horizon latency:* Matched-backbone controls confirm LayerNorm tail compression and barycenter drift reduction while scalar R_{LT} modulation is null, and five-stream horizon sweeps ($S \in \{1..64\}$) retain $2.1\times$ –$4.3\times$ causal Point-F1 gains down to $S = 1$.

3) *Calibration, single-sensor, and slow-ramp boundaries:* Sokal autocorrelation ($\tau_{\text{int}} \leq 206$) leaves 1.5 ($q_{0.995}$) and 0.3 ($q_{0.999}$) expected tail points at $N_{\text{eff}} = 300$. An optional channel branch lifts injected single-sensor ER to 80.7%–96.5% at 6.8%–19.4% unmodified FPR_{test}, and excess causal detection vanishes on slow ramps once $L \geq 1024$ ($4C$); pairing latent panels with raw-level trend charts and testing natural sensor failures remain future priorities.

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